

SUPPLY CHAIN BOTTLENECK AND RECOMMENDATIONS

Prepared By: A Gurukiran

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Candidate Unique no: CA12202582281

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1.INTRODUCTION

This report presents a detailed analysis of the supply chain performance using Power BI dashboards developed from product, supplier, inventory, logistics, and quality management data. The objective of this analysis is to identify operational bottlenecks affecting supply chain performance and provide actionable recommendations to improve efficiency, reduce risks, and support data-driven decision-making.

The supply chain generated total revenue of ₹577.60K and sales volume of 46K units. However, the overall Supply Chain Health Score is 49.57, indicating significant operational improvement opportunities.

2.EXECUTIVE SUMMARY

Key Performance Indicators

- Revenue Generated: ₹577.60K
- Sales Volume: 46K Units
- Supply Chain Health Score: 49.57
- Inventory Coverage Days: 3.11
- Average Defect Rate: 2.28%
- Average Logistics Cost: ₹529.25
- Best Product Category: Skincare
- Supplier Health Score: 70.43

Major bottlenecks identified include:

- Critical inventory shortages
- Overstock inventory accumulation
- Supplier performance inconsistencies
- High quality inspection failures and pending inspections
- Logistics cost inefficiencies

These bottlenecks collectively impact inventory availability, operational costs, service levels, and overall supply chain performance.

3.BOTTLENECK 1: CRITICAL INVENTORY SHORTAGES

Findings

The inventory analysis identified 27 SKUs operating under critical stock conditions.

Key Metrics:

- Average Stock Level: 8.52 Units
- Coverage Days: 0.55
- Demand-to-Stock Ratio: 54.64
- Revenue Exposure: ₹172.90K

Several high-demand products operate with extremely low inventory levels despite generating significant revenue.

Key Insights

- Inventory coverage of only 0.55 days indicates imminent stockout risk.
- Customer demand significantly exceeds available inventory.
- High-revenue SKUs are vulnerable to supply interruptions.
- Fast-moving products dominate the critical inventory category.

Business Impact

- Increased probability of stockouts.
- Lost revenue opportunities.
- Reduced customer satisfaction.
- Potential market share loss.
- Emergency procurement costs.
- Increased pressure on suppliers and logistics operations.

Recommendations

- Prioritize replenishment of all critical inventory SKUs.
- Increase safety stock levels for high-demand products.
- Implement automatic reorder point systems.
- Improve demand forecasting models.
- Develop inventory monitoring alerts.
- Reduce replenishment lead times.

4.BOTTLENECK 2: INVENTORY ALLOCATION IMBALANCE

Findings

The organization simultaneously experiences critical stock shortages and overstock situations.

Inventory Categories:

Critical Stock Revenue: ₹172.90K

Low Stock Revenue: ₹141.28K

Optimal Stock Revenue: ₹131.96K

Overstock Revenue: ₹131.47K

Overstock Analysis:

- 27 SKUs classified as overstocked.
- Average Stock Level: 88.37 Units.
- Coverage Days: 6.03.

Key Insights

- Inventory is not distributed according to demand patterns.
- Excess inventory exists alongside severe shortages.
- Working capital remains locked in slow-moving products.

Business Impact

- Higher inventory carrying costs.
- Increased warehousing expenses.

- Reduced inventory turnover.
- Lower return on inventory investment.

Recommendations

- Rebalance inventory across product categories.
- Introduce ABC inventory classification.
- Monitor inventory turnover regularly.
- Reduce procurement of slow-moving products.
- Implement inventory optimization models.
- Establish category-specific inventory targets.

5.BOTTLENECK 3: SUPPLIER PERFORMANCE VARIABILITY

Findings

Supplier Health Score Analysis:

Supplier 1: 73

Supplier 4: 71

Supplier 2: 71

Supplier 3: 69

Supplier 5: 67

Supplier performance varies considerably across lead times, defect rates, and production capacities.

Key Supplier Metrics

Supplier 1

- Revenue Contribution: ₹157.53K
- Lead Time: 12.59 Days
- Defect Rate: 1.80%

Key Insights

- Supplier 1 demonstrates superior performance.
- Supplier 5 presents the highest operational risk.
- Lead-time variability affects inventory planning.
- Quality inconsistencies increase supply chain uncertainty.

Business Impact

- Supply disruptions.
- Inventory shortages.
- Increased procurement risk.
- Higher quality management costs.
- Reduced service reliability.

Recommendations

- Increase sourcing allocation to Supplier 1.
- Utilize Supplier 4's production capacity more effectively.
- Develop supplier improvement programs.

- Introduce supplier performance scorecards.
- Establish supplier quality benchmarks.
- Implement quarterly supplier reviews.

6.BOTTLENECK 4: QUALITY MANAGEMENT ISSUES

Findings

Average Defect Rate: 2.28%

Quality Inspection Results:

• Pass: 23% • Fail: 36% • Pending: 41%

Haircare products exhibit the highest defect rate among all categories.

Key Insights

- Large number of inspections remain unresolved.
- Quality issues originate from multiple suppliers.
- Defect rates vary across product categories.

Business Impact

- Customer complaints.
- Product returns.
- Additional inspection costs.
- Reduced brand reputation.
- Increased operational inefficiencies.

Recommendations

- Reduce pending inspections immediately.
- Strengthen supplier quality audits.
- Implement root-cause analysis for recurring defects.
- Establish preventive quality controls.
- Create supplier-specific quality improvement plans.
- Target defect rate below 2%.

7.BOTTLENECK 5: LOGISTICS COST INEFFICIENCIES

Findings

Carrier Performance Analysis

Carrier A

- Transportation Cost: ₹497.40
- Revenue Shipped: ₹142.63K
- Average Shipping Time: 6.14 Days

Carrier B

- Transportation Cost: ₹528.50
- Revenue Shipped: ₹250.09K
- Average Shipping Time: 5.30 Days

Carrier C

- Transportation Cost: ₹561.11
- Revenue Shipped: ₹184.88K
- Average Shipping Time: 6.03 Days

Key Insights

- Carrier B provides the best operational balance.
- Carrier A is the most cost-efficient option.
- Carrier C incurs the highest transportation costs.

Business Impact

- Increased logistics expenditure.
- Reduced profitability.
- Inefficient transportation utilization.
- Higher distribution costs.

Recommendations

- Shift shipments from Carrier C to Carrier B where feasible.
- Optimize transportation routes.
- Consolidate shipments to reduce costs.
- Improve shipment planning accuracy.
- Review carrier contracts regularly.
- Monitor logistics performance through KPIs.

8.PRODUCT CATEGORY ANALYSIS

Skincare

- Highest revenue contributor at ₹241.63K.
- Highest sales volume at 21K units.
- Strong demand but lower inventory coverage.

Haircare

- Revenue of ₹174.46K.
- Highest defect rate at 2.48%.
- Requires quality improvement initiatives.

Cosmetics

- Revenue of ₹161.52K.
- Lowest defect rate at 1.92%.
- Best operationally balanced category.

Key Insight

Skincare drives revenue growth, Cosmetics demonstrates operational excellence, and Haircare requires quality improvements.

9. STRATEGIC RECOMMENDATIONS

Based on the analysis, the following strategic initiatives are recommended:

1. Reduce Critical Inventory Risk

Currently, 27 SKUs operate under Critical Stock conditions, with inventory coverage of only 0.55 days and a Demand-to-Stock Ratio of 54.64. Immediate replenishment and safety stock implementation should be prioritized to reduce stockout risk and protect ₹172.90K revenue exposure.

2. Improve Inventory Coverage

Increase average Inventory Coverage Days from 3.11 days to at least 10 days through improved demand forecasting and inventory planning. This will improve product availability and service levels.

3. Rebalance Inventory Allocation

The business simultaneously maintains 27 Critical Stock SKUs and 27 Overstocked SKUs, indicating inefficient inventory allocation. Excess inventory should be reallocated toward high-demand products to improve inventory utilization.

4. Increase Procurement from High-Performing Suppliers

Supplier 1 achieved the highest Supplier Health Score (73) with the lowest defect rate (1.80%) and shortest lead time (12.59 days). Procurement allocation should be increased toward Supplier 1 to improve supply reliability.

5. Improve Supplier 5 Performance

Supplier 5 recorded the lowest Health Score (67), highest lead time (16.33 days) and highest defect rate (2.67%). A supplier improvement program should be implemented to reduce operational risk.

6. Reduce Product Defects

Current Quality Defect Rate stands at 2.28%, with Haircare showing the highest defect rate (2.48%). Quality audits and root-cause analysis should target defect reduction below 2%.

7. Clear Pending Quality Inspections

Approximately 41% of inspections remain pending, delaying inventory release and increasing operational uncertainty. The target should be to reduce pending inspections below 10%.

8. Optimize Logistics Costs

Carrier C incurs the highest logistics cost (₹561.11) compared to Carrier A (₹497.40). Shipment allocation and route optimization should be reviewed to achieve at least 10% logistics cost reduction.

9. Focus on High-Revenue Categories

Skincare contributes ₹241.63K (42% of total revenue) and should receive priority inventory allocation, demand forecasting, and replenishment planning to prevent revenue loss.

10. Improve Supply Chain Health Score

Implementing the above initiatives can increase the Supply Chain Health Score from 49.57 to a target of 80+, resulting in improved inventory availability, reduced costs, enhanced supplier performance, and higher customer satisfaction.

10.CONCLUSION

The analysis reveals that the primary challenge facing the organization is not demand generation but operational execution. Strong revenue generation and product demand indicate healthy market performance; however, inventory shortages, supplier variability, logistics inefficiencies, and quality issues significantly reduce overall supply chain effectiveness.

The most critical bottleneck is inventory management, particularly the presence of high-revenue products operating under critical stock conditions. Addressing inventory shortages, strengthening supplier performance, improving quality control processes, and optimizing logistics operations will significantly enhance service levels, reduce costs, and improve overall supply chain resilience.

Implementation of the recommended actions is expected to improve inventory availability, increase operational efficiency, reduce risk exposure, and elevate the Supply Chain Health Score toward best-in-class performance levels.